

Fundamentals of Physical Geography

UNIT	KEY CONCEPTS	LEARNING OUTCOME	ACTIVITY / RESOURCE
1. Geography as a discipline	- Nature and Scope - Science & Spatial attributes - Branches of Geography - Importance of physical geography (Nature and scope) Reasons : Added for the initial knowledge of Geography.	- Nature and scope is important in the subject study. - Importance of Geography and the knowledge of its field etc.	Discussion on Geography as a discipline through Charts and diagram.
2. Earth	- Origin and Evolution of the Earth. - Interior of the Earth. - Wegener's continental drift theory and plate tectonics. - Geological process : Earthquake, volcanoes.	Earth and its related fact ex. its structure and knowledge about its happening phenomena.	To prepare a model of the interior of the earth.
3. Landforms	- Minerals and Rocks - Types of Rocks and its characteristics. - Geomorphic process - weathering and erosion. - Landforms and its types.	- To give knowledge about the changes that take place in the physiography of the earth. - Also to impart knowledge about soil formation.	To give knowledge about landforms of the nearest places through field survey / short tour. To show areas of soil
4. Climate	- Composition and structure - Elements of Weather and Climate. - Insolation : Angle of incidence of Sun's ray & distribution. - Heat Budget - Heating and Cooling of Atmosphere. - Conduction, Convection, Radiation. - Problems of Global Warming. - Green House Effects. - Pressure Belts. Winds– Planetary, Seasonal & Local and its effects. - Air masses and fronts. - Cyclones : Tropical & Temperate. - Condensation and Evaporation process.	- To impart knowledge of the Atmosphere and its weather and Climate. - Effects of changing weather phenomena on human beings.	- To give knowledge of weather phenomena's through observations and experiments. - Discussion on Geography as a discipline through Charts and diagram.

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	• Forms of condensation : Dew, frost, fog, mist & clouds. • Precipitation. • Types of Rainfall.		
5. Hydrosphere (Water) oceans	• Hydrological Cycle • Study of Submarine Relief. • Distribution of temperature and Salinity. • Movements of Ocean water-waves, tides and currents.	• Hydrosphere is an important resource on Earth. Therefore its knowledge is necessary.	• Through Maps and diagrams.
6. Biosphere	• Ecosystems • Role of Man in Ecological Imbalance. • Human and Environmental effects.	• To give knowledge of Biosystem which gives knowledge of physical system also at this level. • To give knowledge of various aspects of Economic Geography.	• To prepare a Model Ecosystem.

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Economic Geography

UNIT	KEY CONCEPTS	LEARNING OUTCOME	ACTIVITY / RESOURCE
1. Resource	• Meaning and Scope. • Concept • Classification and Conservation.	• Information about Resources.	• Charts & Models.
2. Man and Environment	• Major natural Regions of the world : - Equatorial Region - Monsoon Region - Tropical Region - Temperate grassland. • Human Life	• Information about Economic activity. • To inform about the consequences on Human life and Economy of Economic activities.	• Usage of Maps and Globe.
3. Main crops of the World	• Grains – Rice, Wheat & Pulses. • Cash Crops – Cotton & Sugarcane. • Plantation Crops : Tea • Milk Production • Geographical conditions, distribution and world-trade of different crop's production.	• To make understand, the need of grains / crops in the daily life. • To make understand the effect of agriculture on World's economy.	• Maps • Project works • Diagrams
4. Major industries of the World	• Utility and distribution of Iron-ore, Cotton-textile. • Factors of Industrialization and its effects on the economy.	• Knowledge of Major Industries of the World and its effect on the economy.	• Maps and Diagrams.

Practical Assignment

UNIT	KEY CONCEPTS	LEARNING OUTCOME	ACTIVITY / RESOURCE
1. Basics of Map (Fundamentals of Map)	- Types of Maps and Scales. - Finding direction on Map. - Use of Symbols (Conventional) - Projections – Types and Utility. - Conical projection with 1 standard parallel.	Aim of Map study is to bring the landforms and its features on a plain paper and to provide knowledge of its study.	- Line formation. - To show some Major places and landforms and world map.
2. Study of Maps Related to Landforms and Weather Phenomena (TOPO SHEET)	- Study of topographical-sheet (Topo-sheet) - Toposheet of Bihar, Based on R : f 1:50,000 represented by A.S.I. (Archaeological Survey of India).	To give knowledge of landforms through maps.	- Topo-sheet - Maps - Conventional symbols
3. Contour Lines	- Profiles on the basis of Contour lines. - Mountains, Plateau, Valleys, Waterfall & Slope.	Study of converse topography on plain land and its knowledge.	- Maps - Toposheet
4. Utility of Weather Related Instruments	- Thermometer - Barometer - Rain-gauge - Readings of Instruments	Readings of Instrument necessary	To note the temperature of classroom and to discuss its differences.
5. Practical Records	- Records - Practical books	Serial maintenance and representation of Practical records.	- Survey, field survey, - Picturisation of Practical works.

